

The Perception Gap: 911 Calls and Official Police Records

This project explores the perception gap between police dispatch calls and confirmed crime reports in Montgomery County by analyzing how incidents are initially perceived through 911 calls and later classified in official police crime records. This topic is important because police departments allocate time, personnel, and operational resources largely based on incoming dispatch demand. Even incidents that do not become official crimes still require officer response and investigation. Understanding the perception gap therefore provides insight into public behavior, police workload, operational efficiency, and community safety concerns. The goal of this project was to measure the perception gap between police dispatch activity and confirmed crime outcomes by examining how often dispatch calls did not lead to confirmed crime reports and whether these patterns varied across districts and over time.

Datasets Used

The Crime dataset contains official police crime reports recorded by Montgomery County law enforcement, including information about crime categories, severity levels, police districts, and date/time variables. The Police Dispatch Incidents dataset contains records of incidents reported to police dispatch services and represents the initial public perception of events before formal police investigation.

Tools Used

Most of the data preparation, statistical analysis, and visualization work was completed in R using packages such as tidyverse, lubridate, and ggplot2, while Tableau was used to create the interactive dashboard. GitHub was used for project organization and documentation storage.

Data Cleaning and Pre Processing

First, all column names were standardized by converting them to lowercase and replacing spaces with underscores. Next, the date and time variables were formatted, and time related variables were created, including year, month, hour of the day, and time period. One of the most important preprocessing steps was the creation of a severity ranking system. Crimes were grouped into four standardized levels ranging from non criminal incidents (0) to serious violent crimes (3). A similar ranking system was also made for dispatch calls in order to classify dispatch incidents into comparable severity categories.

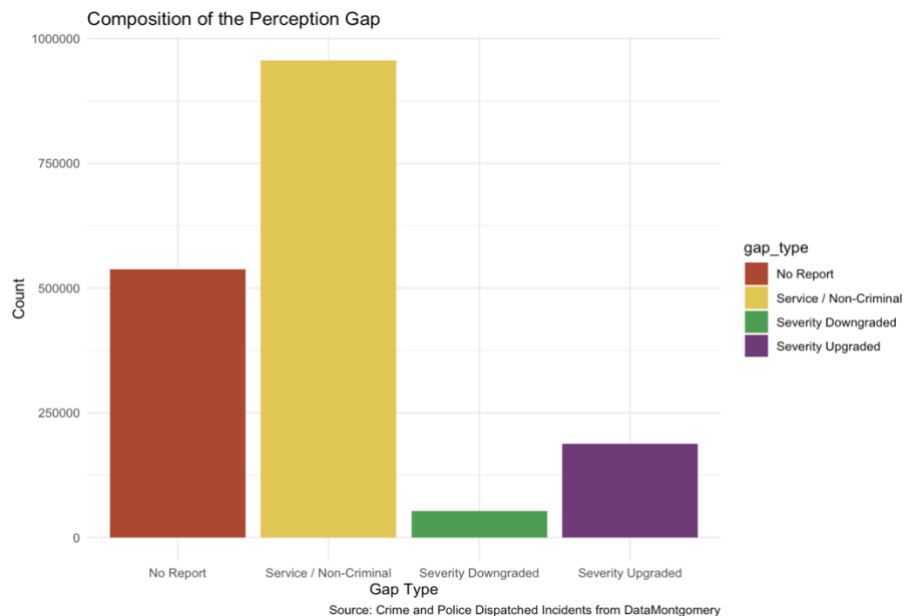
Defining the Perception Gap

Several variables were created to compare dispatch severity levels with final crime outcomes. A “Severity Match” occurs when the dispatch and crime classifications are at the same severity level. “Severity Downgraded” means the dispatch call was initially considered more severe than the final crime report, while “Severity Upgraded” means the final crime report was more severe than the original dispatch call. Other categories included incidents with no official crime report and “Service / Non Criminal” incidents such as welfare checks, disturbances, and administrative calls that required police response without involving confirmed criminal activity.

Exploratory Data Analysis

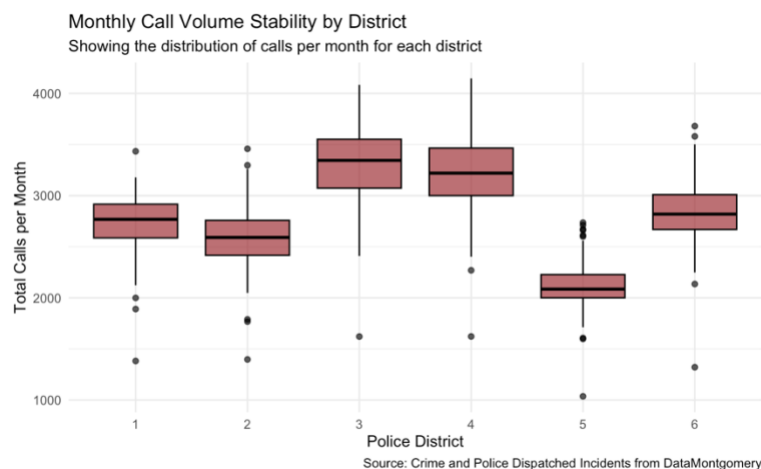
The EDA phase showed that most dispatch calls didn’t lead to confirmed crime reports, indicating that much of police demand is for service calls, welfare checks, disturbances, or non criminal situations. When analyzing only confirmed crimes, Severity Upgraded cases appeared

more frequently than Severity Downgraded cases, suggesting some initially low priority incidents later became serious crimes after investigation.



Perception gap rates were consistently high across all police districts, with Districts 1 and 2 showing higher rates and Districts 5 and 6 showing lower rates. Mismatch rates also varied across districts, with District 2 showing some of the highest mismatch levels.

Call volume was not evenly distributed across the county. Districts 3 and 4 recorded the highest total call volume, while District 5 consistently showed the lowest activity levels.



Statistical Analysis

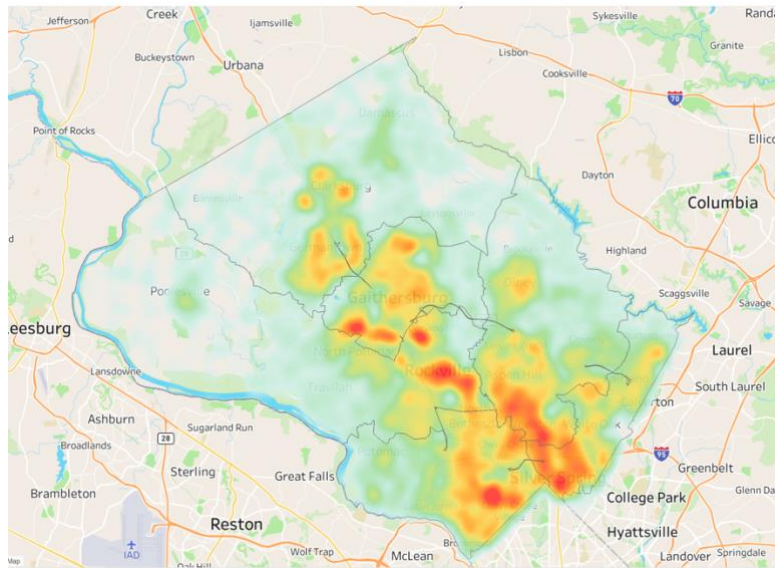
An ANOVA model was used to compare perception gap rates across police districts. The null hypothesis stated that all districts had equal mean perception gap rates, while the alternative hypothesis stated that at least one district differed. The results showed statistically significant differences between districts. Therefore, the null hypothesis was rejected. This means that some districts experience meaningfully different rates than others.

Then, I performed a Tukey post-hoc test to identify which districts differed significantly. The largest difference observed was between district 2 and district 5 (district 5 showed lower perception gap rates).

I also created a multiple linear regression model to see how police district and year affect the total perception gap rate. The model explained approximately 24% of the variation. Although this indicates that district and year are meaningful predictors, it also suggests that additional factors influence the perception gap. Diagnostic plots showed acceptable linearity, slight heteroscedasticity, and some heavy tails and outliers, but no severe violations.

Dashboard and Final Data Product

The dashboard was created to provide an interactive and visual representation of the perception gap analysis across Montgomery County. Using Tableau, the dashboard integrates multiple dimensions of the project, including service versus non service calls, severity mismatch patterns, hourly trends, district comparisons, and spatial distribution mapping. The dashboard allows users to explore how dispatch activity differs from confirmed crime outcomes through dynamic filtering and visualization tools.



Challenges

Preparing and integrating dispatch and crime datasets was challenging due to different formats and classifications. The lack of official crime reports for many dispatch calls further complicated the matching process. Creating a consistent severity ranking system was also hard.

What Was Rewarding

One of the most rewarding aspects of this project was being able to work with real public safety data to analyze meaningful community patterns. It was also rewarding to see the analysis come together visually through the dashboard.

What Could Be Improved

This project could be improved by adding other variables such as demographic and socioeconomic information to provide deeper context for the perception gap patterns. Future versions of the project could also include more advanced statistical or machine learning models.

Recommendations

The findings of this project suggest that Montgomery County police departments may benefit from monitoring districts with higher perception gap rates, particularly Districts 1 and 2, to better understand the factors contributing to these patterns. Continued review of severity classification consistency between dispatch calls and confirmed crime reports may also help improve operational accuracy and data quality. In addition, public information and outreach efforts could help the public better understand how incidents are classified and when emergency reporting is appropriate. Increasing public awareness about the difference between service related incidents and confirmed criminal activity may also help reduce unnecessary dispatch demand while improving communication between the community and public safety agencies.

Acknowledgements

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References

- [GitHub](#)
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- Montgomery County FY2026 Approved Operating Budget (Office of Management and Budget)
- Montgomery County Police Department Press Releases and CitizenPortal Emergency Communications Resources
- DataMontgomery Open Data Portal